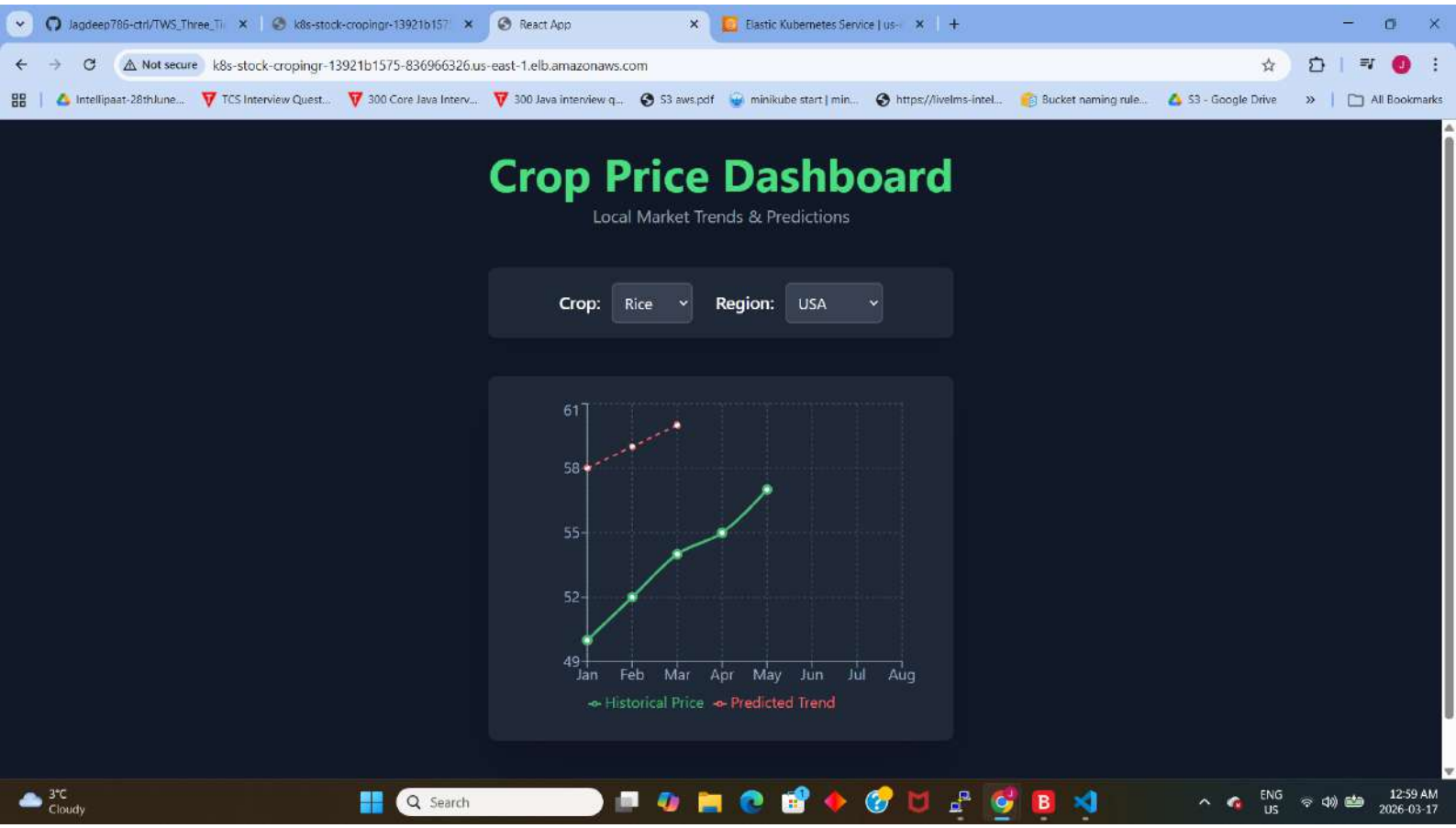
```
┌──────────────┐
│    User      │
└──────────────┘
       │
       ▼
┌──────────────┐
│  Frontend    │
│ (React App)  │
└──────────────┘
       │
       ▼
┌──────────────┐
│  API Gateway │
│(Spring Boot) │
└──────────────┘
       │
   ┌───┴───┬───┴───┬───┴───┐
   ▼       ▼       ▼
┌──────┐ ┌──────┐ ┌──────┐
│Data  │ │Trend │ │Write │
│Service│ │Service│ │Service│
│(Go + │ │(Python│ │(Node.│
│MySQL)│ │ ML)  │ │ js)  │
│Hist- │ │Pre-  │ │POST  │
│orical│ │diction│ │Data  │
└──────┘ └──────┘ └──────┘
   │       │       │
   └───┬───┴───┬───┘
       ▼       ▼
Historical Data   Trend / Predictions
(from DB)         (from ML)
       │
       ▼
API Gateway merges all results
       │
       ▼
Frontend receives combined JSON
       │
       ▼
Dashboard / Charts displayed
```


Jagdeep786-ctrl/TWS_Three_Ti
k8s-stock-cropingr-13921b157
React App
Elastic Kubernetes Service | us-
Not secure
k8s-stock-cropingr-13921b1575-836966326.us-east-1.elb.amazonaws.com
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All Bookmarks
Crop Price Dashboard
Local Market Trends & Predictions
Crop:
Rice
Region:
USA
61
58
55
52
49
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Historical Price
Predicted Trend
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Jagdeep786-ctrl/TWS_Three_Ti | k8s-stock-cropingr-13921b157 | React App | Elastic Kubernetes Service | us-

Not secure k8s-stock-cropingr-13921b1575-836966326.us-east-1.elb.amazonaws.com/api/data?crop=Rice&region=USA

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Pretty-print


```
{
  "trend": {
    "trend": [
      {
        "date": "2026-03-16",
        "price": 57
      },
      {
        "date": "2026-03-21",
        "price": 58.58
      }
    ]
  },
  "historical": [
    {
      "date": "2026-03-03",
      "price": 55.5
    },
    {
      "date": "2026-03-04",
      "price": 55.75
    },
    {
      "date": "2026-03-05",
      "price": 56
    },
    {
      "date": "2026-03-06",
      "price": 55.9
    },
    {
      "date": "2026-03-07",
      "price": 56.2
    },
    {
      "date": "2026-03-08",
      "price": 56.5
    },
    {
      "date": "2026-03-09",
      "price": 56.4
    },
    {
      "date": "2026-03-10"
```


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ubuntu@ip-172-31-92-86: ~/crop_dashboard

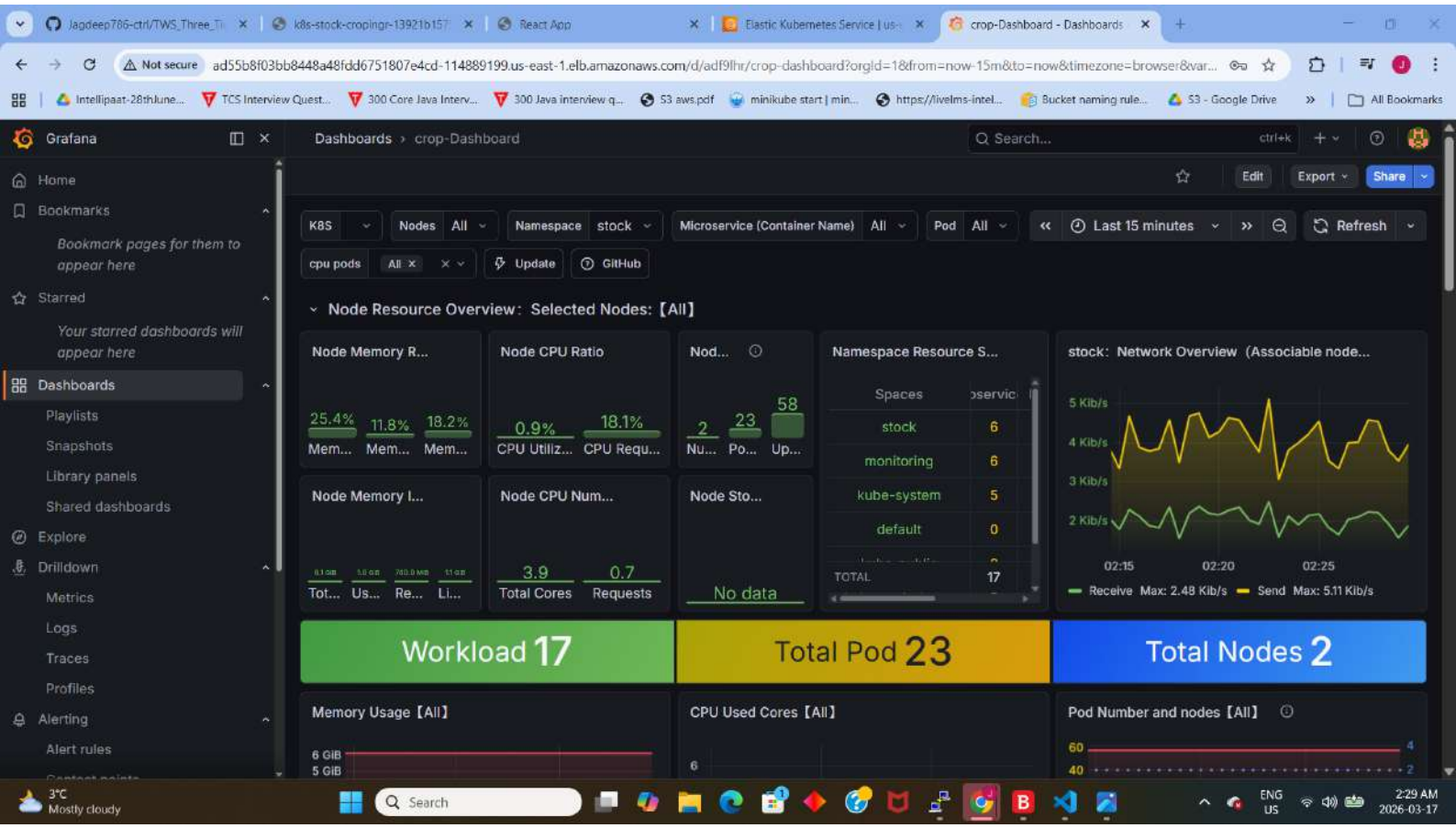
```
ubuntu@ip-172-31-92-86:~/crop_dashboard$ kubectl get pods -n stock
NAME                                  READY   STATUS    RESTARTS   AGE
api-gateway-9b4874874-fpc7f           1/1     Running   0          39m
data-service-7dbc87c7c7-flkcg         1/1     Running   0          17m
frontend-deployment-5cb5798db5-2wd6g  1/1     Running   0          39m
mysql-deployment-67df9f4b66-kv45j     1/1     Running   0          21m
trend-service-78b54f4c57-2jsxx        1/1     Running   0          39m
writer-service-bfc5879c-l4gvb         1/1     Running   0          39m
ubuntu@ip-172-31-92-86:~/crop_dashboard$ kubectl get svc -n stock
NAME                  TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)
 AGE
api-gateway           ClusterIP   10.100.245.176   <none>        8080/TCP
 40m
data-service-go       ClusterIP   10.100.32.5      <none>        8081/TCP
 40m
frontend-service      NodePort    10.100.125.47    <none>        80:30001/TCP
 40m
mysql-db              ClusterIP   10.100.21.54     <none>        3306/TCP
 40m
trend-service-python  ClusterIP   10.100.5.28      <none>        5000/TCP
 40m
writer-service        ClusterIP   10.100.18.60     <none>        3001/TCP
 40m
ubuntu@ip-172-31-92-86:~/crop_dashboard$
```

ubuntu@ip-172-31-92-86: ~/crop_dashboard

```
ubuntu@ip-172-31-92-86:~/crop_dashboard$ kubectl get ingress -n stock
NAME           CLASS    HOSTS   ADDRESS
                      PORTS   AGE
crop-ingress   alb      *       k8s-stock-cropingr-13921b1575-836966326.us-east-1
.elb.amazonaws.com   80      41m
ubuntu@ip-172-31-92-86:~/crop_dashboard$
```

ubuntu@ip-172-31-92-86: ~/crop_dashboard

```
ubuntu@ip-172-31-92-86:~/crop_dashboard$ kubectl logs mysql-deployment-67df9f4b66-kv45j -n stock
2026-03-17 06:41:03+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.45-1.el9 started.
2026-03-17 06:41:03+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
2026-03-17 06:41:03+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.45-1.el9 started.
2026-03-17 06:41:03+00:00 [Note] [Entrypoint]: Initializing database files
2026-03-17T06:41:03.637127Z 0 [Warning] [MY-011068] [Server] The syntax '--skip-host-cache' is deprecated and will be removed in a future release. Please use SET GLOBAL host_cache_size=0 in
stead.
2026-03-17T06:41:03.637196Z 0 [System] [MY-013169] [Server] /usr/sbin/mysqld (mysqld 8.0.45) initializing of server in progress as process 80
2026-03-17T06:41:03.643268Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
2026-03-17T06:41:04.308629Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
2026-03-17T06:41:05.400553Z 6 [Warning] [MY-010453] [Server] root@localhost is created with an empty password ! Please consider switching off the --initialize-insecure option.
2026-03-17 06:41:08+00:00 [Note] [Entrypoint]: Database files initialized
2026-03-17 06:41:08+00:00 [Note] [Entrypoint]: Starting temporary server
2026-03-17T06:41:08.366791Z 0 [Warning] [MY-011068] [Server] The syntax '--skip-host-cache' is deprecated and will be removed in a future release. Please use SET GLOBAL host_cache_size=0 in
stead.
2026-03-17T06:41:08.368033Z 0 [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 8.0.45) starting as process 122
2026-03-17T06:41:08.383609Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
2026-03-17T06:41:08.663882Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
```


Jagdeep786-ctrl/TWS_Three_Ti
k8s-stock-cropingr-13921b157
React App
Elastic Kubernetes Service | us-
crop-Dashboard - Dashboards
Not secure
ad55b8f03bb8448a48fdd6751807e4cd-114889199.us-east-1.elb.amazonaws.com/d/adf9lhr/crop-dashboard?orgId=1&from=now-15m&to=now&timezone=browser&var...
Intellipaat-28thJune...
TCS Interview Quest...
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Bucket naming rule...
S3 - Google Drive
All Bookmarks
Grafana
Dashboards › crop-Dashboard
Search...
ctrl+k
Home
Bookmarks
Bookmark pages for them to appear here
Starred
Your starred dashboards will appear here
Dashboards
Playlists
Snapshots
Library panels
Shared dashboards
Explore
Drilldown
Metrics
Logs
Traces
Profiles
Alerting
Alert rules
Edit
Export
Share
K8S
Nodes All
Namespace stock
Microservice (Container Name) All
Pod All
Last 15 minutes
Refresh
cpu pods All
Update
GitHub
Node Resource Overview: Selected Nodes: 【All】
Node Memory R...
25.4% Mem...
11.8% Mem...
18.2% Mem...
Node CPU Ratio
0.9% CPU Utiliz...
18.1% CPU Requ...
Nod...
2 Nu...
23 Po...
58 Up...
Namespace Resource S...
Spaces
stock 6
monitoring 6
kube-system 5
default 0
TOTAL 17
stock: Network Overview (Associable node...
5 Kib/s
4 Kib/s
3 Kib/s
2 Kib/s
02:15
02:20
02:25
Receive Max: 2.48 Kib/s
Send Max: 5.11 Kib/s
Node Memory I...
Tot... Us... Re... Li...
Node CPU Num...
3.9 Total Cores
0.7 Requests
Node Sto...
No data
Workload 17
Total Pod 23
Total Nodes 2
Memory Usage 【All】
6 GiB
5 GiB
CPU Used Cores 【All】
6
Pod Number and nodes 【All】
60
40
4
2
3°C
Mostly cloudy
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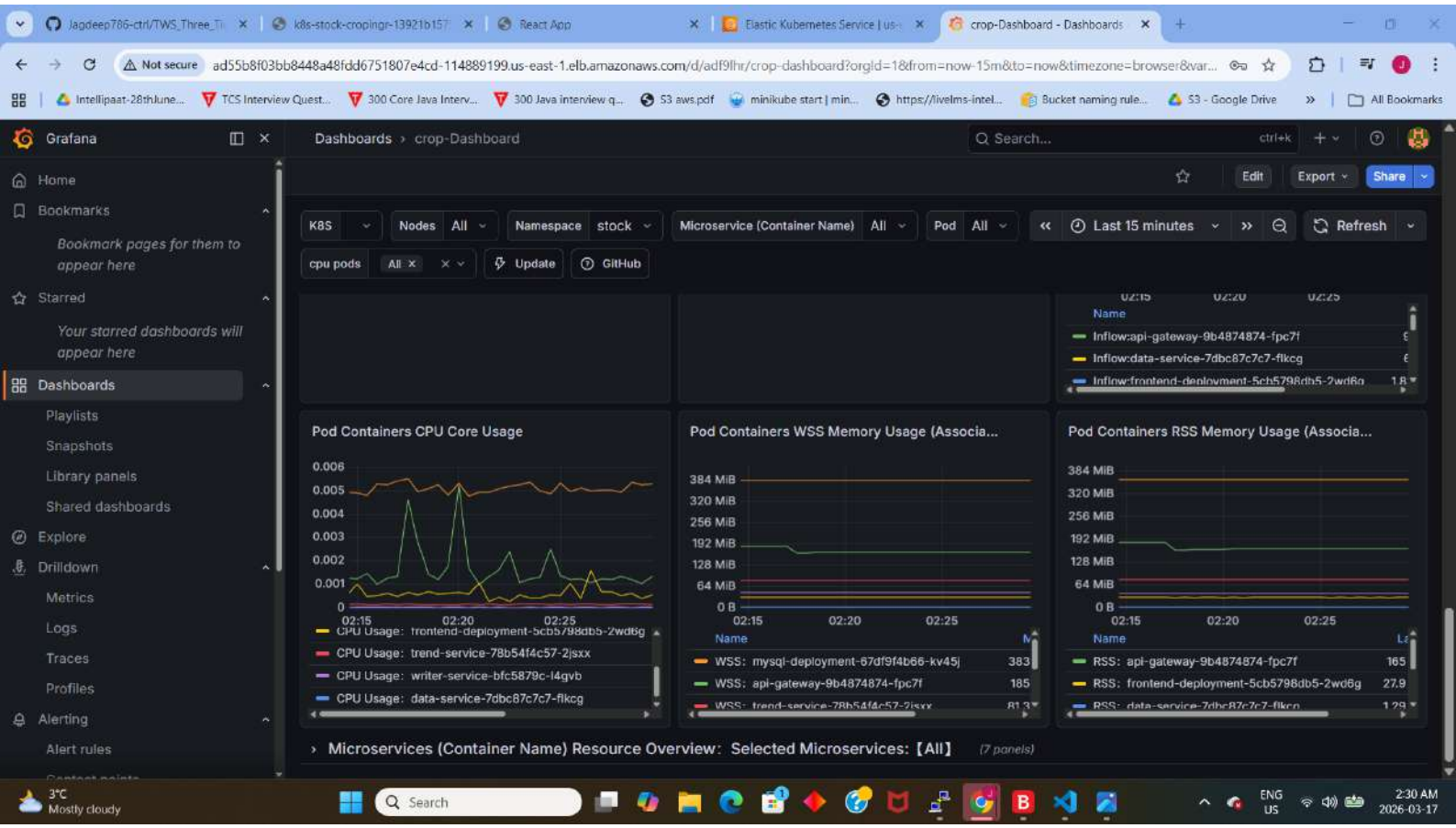
2026-03-17


crop-Dashboard - Dashboards
Not secure
ad55b8f03bb8448a48fdd6751807e4cd-114889199.us-east-1.elb.amazonaws.com/d/adf9lhr/crop-dashboard?orgId=1&from=now-15m&to=now&timezone=browser&var...
Grafana
Dashboards › crop-Dashboard
Search...
Home
Bookmarks
Bookmark pages for them to appear here
Starred
Your starred dashboards will appear here
Dashboards
Playlists
Snapshots
Library panels
Shared dashboards
Explore
Drilldown
Metrics
Logs
Traces
Profiles
Alerting
Alert rules
Edit
Export
Share
K8S
Nodes All
Namespace stock
Microservice (Container Name) All
Pod All
Last 15 minutes
Refresh
cpu pods All
Update
GitHub
Pod Containers CPU Core Usage
CPU Usage: frontend-deployment-5cb5798db5-2wd6g
CPU Usage: trend-service-78b54f4c57-2jsxx
CPU Usage: writer-service-bfc5879c-l4gvb
CPU Usage: data-service-7dbc87c7c7-flkcg
Pod Containers WSS Memory Usage (Associa...
WSS: mysql-deployment-67df9f4b66-kv45j 383
WSS: api-gateway-9b4874874-fpc7f 185
Pod Containers RSS Memory Usage (Associa...
RSS: api-gateway-9b4874874-fpc7f 165
RSS: frontend-deployment-5cb5798db5-2wd6g 27.9
Inflow:api-gateway-9b4874874-fpc7f
Inflow:data-service-7dbc87c7c7-flkcg
Microservices (Container Name) Resource Overview: Selected Microservices: 【All】 (7 panels)
3°C Mostly cloudy
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Jagdeep786-ctrl/TWS_Three_Ti | k8s-stock-cropingr-13921b157 | React App | Instances | EC2 | us-east-1 | crop-Dashboard - Dashboards

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:instanceState=running

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aws | Search [Alt+S] | United States (N. Virginia) | jagdeepsodhi264@gmail.com (2101-9073-2631) | jagdeepsodhi264@gmail.com

EC2 > Instances

EC2

- Dashboard
- AWS Global View
- Events
- **Instances**
  - Instances
  - Instance Types
  - Launch Templates
  - Spot Requests
  - Savings Plans
  - Reserved Instances
  - Dedicated Hosts
  - Capacity Reservations
  - Capacity Manager New
- **Images**
  - AMIs
  - AMI Catalog
- **Elastic Block Store**

Instances (3) Info

Connect | Instance state | Actions | Launch instances

Find Instance by attribute or tag (case-sensitive) | All states

Instance state = running | Clear filters

1

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
crop-dashboar...	i-098f3429978591a60	Running	c7i-flex.large	3/3 checks passed	View alarms +	us-east-1c	ec2-34-229
crop-dashboar...	i-0d45a25ab788b258a	Running	c7i-flex.large	3/3 checks passed	View alarms +	us-east-1a	ec2-100-53
crop_dashboard	i-07dfb70aee72f0c82	Running	c7i-flex.large	3/3 checks passed	View alarms +	us-east-1b	ec2-13-221

Select an instance

CloudShell | Feedback | Console Mobile App

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Jagdeep786-ctrl/TWS_Three_Ti | k8s-stock-cropingr-13921b157 | React App | Elastic Kubernetes Service | us- | crop-Dashboard - Dashboards

us-east-1.console.aws.amazon.com/eks/clusters/crop-dashboard-newcluster?region=us-east-1&selectedTab=cluster-networking-tab

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Amazon Elastic Kubernetes Service > Clusters > crop-dashboard-newcluster

Amazon Elastic Kubernetes Service

- Dashboard
- Clusters
- **Settings**
  - Dashboard settings
  - Console settings
- **Amazon EKS Anywhere**
  - Enterprise Subscriptions
- **Related services**
  - Amazon ECR
  - AWS Batch

Documentation

EKS workshops New

crop-dashboard-newcluster

Delete cluster | Upgrade version | Monitor cluster

End of standard support for Kubernetes version 1.34 is December 1, 2026. Upgrade

Cluster info Info

Status	Kubernetes version Info	Support period	Provider
Active	1.34	Standard support until December 1, 2026	EKS

Cluster health	Upgrade insights	Node health issues	Capability issues
0	5	0	0

Overview | Resources | Compute | Networking | Add-ons 1 | Capabilities | Access | Observability | Update history &

Networking

Manage

VPC Info	Subnets	Cluster security group Info	API server endpoint access Info
vpc-0d72f7cf5f63e963e	subnet-0abaa0f29cdf49dea subnet-0596f16071a0ca326	sg-0e5c743d53950eebb	Public

Cluster IP address family Info | Additional security groups | Public access source allowlist

CloudShell | Feedback | Console Mobile App

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Jagdeep786-ctrl/TWS_Three_Ti | k8s-stock-cropingr-13921b157 | React App | Load balancer details | EC2 | us- | crop-Dashboard - Dashboards

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LoadBalancer:loadBalancerArn=arn:aws:elasticloadbalancing:us-east-1:210190732631:loadbalancer/app/k8s-stock-c...

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EC2 > Load balancers > k8s-stock-cropingr-13921b1575

- Capacity Reservations
- Capacity Manager New
- **Images**
  - AMIs
  - AMI Catalog
- **Elastic Block Store**
  - Volumes
  - Snapshots
  - Lifecycle Manager
- **Network & Security**
  - Security Groups
  - Elastic IPs
  - Placement Groups
  - Key Pairs
  - Network Interfaces
- **Load Balancing**
  - Load Balancers
  - Target Groups
  - Trust Stores
- **Auto Scaling**
  - Auto Scaling Groups

k8s-stock-cropingr-13921b1575

Actions

Details

Load balancer type	Status	VPC	Load balancer IP address type
Application	Active	vpc-0d72f7cf5f63e963e	IPv4
Scheme	**Hosted zone**	**Availability Zones**	**Date created**
Internet-facing	Z35SXDOTRQ7X7K	subnet-0abaa0f29cdf49dea us-east-1a (use1-az1) subnet-0596f16071a0ca326 us-east-1c (use1-az4)	March 17, 2026, 00:27 (UTC-06:00)

Load balancer ARN
arn:aws:elasticloadbalancing:us-east-1:210190732631:loadbalancer/app/k8s-stock-cropingr-13921b1575/2e68af1086600b3f

DNS name Info
k8s-stock-cropingr-13921b1575-836966326.us-east-1.elb.amazonaws.com (A Record)

Listeners and rules | Network mapping | Resource map | Security | **Monitoring** | Integrations | Attributes | Capacity | Tags

Investigate with AI - *new* | 1h | 3h | 12h | 1d | 3d | 1w | Custom | UTC timezone | Explore related

Target Response Time — Seconds — 96.5 ms — 48.2 ms

Requests — Count — 165 — 82.5

Rule Evaluations — Count — 318 — 159

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